

7.2 Canonical Correlation Analysis

Find paired directions that co-vary across two views.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

Figures and notes follow Deisenroth, Faisal, and Ong, *Mathematics for Machine Learning*; Theodoridis; and Géron.

1. Canonical correlation analysis (CCA)

PCA reduces dimension for **one** data set. Often you have **several** related sets from different sources.

Canonical correlation analysis (CCA) processes **two** sets together. The correlation of two random vectors depends on the coordinate system you write them in.

CCA looks for a pair of **linear** maps, one per set, so that the transformed variables are **maximally correlated**.

Two objects to keep straight:

- **Canonical variables** (variates)
- **Canonical correlation**

2. The math

Two random vectors $\mathbf{x} \in \mathbb{R}^p$ and $\mathbf{y} \in \mathbb{R}^q$, observations x_n, y_n for $n = 1, \dots, N$.

As in PCA, start with one pair of directions, $u_{x,1}$ and $u_{y,1}$, that maximize the correlation of the projections.

Equivalently, the constrained problem

$$\max_{u_x, u_y} u_x^\top \hat{\Sigma}_{x,y} u_y \quad \text{subject to} \quad u_x^\top \hat{\Sigma}_{x,x} u_x = 1, \quad u_y^\top \hat{\Sigma}_{y,y} u_y = 1,$$

where $\Sigma_{x,y}$ is the **cross-covariance**.

Compared with PCA: two directions, not one; the constraints use a **weighted Σ norm**, not the Euclidean norm; the objective is **correlation of the two projections**, not variance of one.

Lagrangian:

$$\mathcal{L}(u_x, u_y, \lambda_x, \lambda_y) = u_x^\top \hat{\Sigma}_{x,y} u_y - \frac{\lambda_x}{2} (u_x^\top \hat{\Sigma}_{x,x} u_x - 1) - \frac{\lambda_y}{2} (u_y^\top \hat{\Sigma}_{y,y} u_y - 1).$$

Gradients in u_x and u_y set to zero give $\lambda_x = \lambda_y = \lambda$ and

$$\hat{\Sigma}_{x,y} u_y = \lambda \hat{\Sigma}_{x,x} u_x, \quad \hat{\Sigma}_{y,x} u_x = \lambda \hat{\Sigma}_{y,y} u_y.$$

Solve the second for u_y (assuming $\hat{\Sigma}_{y,y}$ invertible) and substitute:

$$\hat{\Sigma}_{x,y} \hat{\Sigma}_{y,y}^{-1} \hat{\Sigma}_{y,x} u_x = \lambda^2 \hat{\Sigma}_{x,x} u_x$$

and

$$u_y = \frac{1}{\lambda} \hat{\Sigma}_{y,y}^{-1} \hat{\Sigma}_{y,x} u_x.$$

If $\hat{\Sigma}_{x,x}$ is invertible too, this is an ordinary eigenproblem:

$$(\hat{\Sigma}_{x,x}^{-1} \hat{\Sigma}_{x,y} \hat{\Sigma}_{y,y}^{-1} \hat{\Sigma}_{y,x}) u_x = \lambda^2 u_x.$$

So $u_{x,1}$ is an eigenvector of that product. Take the eigenvector for the **largest** eigenvalue λ^2 to maximize the correlation.

$u_{x,1}$ and $u_{y,1}$ are the **normalized canonical correlation basis vectors**. The eigenvalue λ^2 is the **squared canonical correlation**. The projections $u_{x,1}^\top \mathbf{x}$ and $u_{y,1}^\top \mathbf{y}$ are the **canonical variates**.

3. What to read off in practice

Two questions after you fit CCA:

- **Canonical correlation.** How much is each canonical variable from set 1 correlated with its counterpart in set 2?
 - **Canonical variates.** Which original variables does each variate represent?
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Practice

1. CCA needs two views of the same objects. Name a pair.
2. What is CCA maximizing that PCA is not, and what replaces the unit-Euclidean constraint?
3. After you compute $u_{x,1}$, how do you get $u_{y,1}$?